LEONARDO VANNI

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EDUCATION

Technische Universität München, Munich, Germany — Oct 2026 – Sep 2028 (expected)
Incoming M.Sc. in Mathematics in Science and Engineering

- Application area: robotics perception, including multiple-view geometry, 3D geometry learning, and deep learning for robotics.

Bocconi University, Milan, Italy — Sep 2023 – Jul 2026
B.Sc. in Mathematics and Computing Sciences for Artificial Intelligence

- Final grade: 110/110 cum laude.
- Exchange semester, **École Polytechnique**, Palaiseau, France — Sep 2025 – Jan 2026
  GPA: 4.3/4.0 (A+).

CURRENT RESEARCH

Dynamic 3D reconstruction and reliable geometric learning — Dec 2025 – Present
Computer Vision Researcher, Bocconi University

- Research on reconstructing dynamic and egocentric scenes as their structure changes, and on estimating when learned 3D geometry can be trusted.
- In collaboration with researchers at Bocconi, TUM, and Google. Details available on request.

RESEARCH EXPERIENCE

Geometrically-Grounded Uncertainty Quantification for Foundational 3D Vision Models — 2026
B.Sc. thesis, Bocconi University; Supervisor: Alessandro Pigati

- Extended VGGT with a lightweight branch that predicts full 6×6 camera-pose covariances on $\mathrm{SE}(3)$ while preserving the frozen mean-pose pathway.
- Validated on CO3D and EPIC-KITCHENS: a single temperature calibrated the uncertainty, which matched bundle-adjustment covariance structure and flagged mislocalized frames.

Vision Dental — Jun 2025 – Sep 2025
Computer Vision Researcher

- Built a GPU-accelerated CT reconstruction and 3D U-Net segmentation/classification pipeline for tooth- and lesion-level patient reports.
- Added ensemble-based voxel uncertainty, temperature calibration, and confidence-based abstention for high-risk predictions.

SELECTED TECHNICAL EXPERIENCE

Kornia – *Open-Source Contributor* — Jun 2026 – Present
Two merged pull requests

- Added MAE, RMSE, and bad-pixel stereo disparity metrics with tests and documentation; removed import-time TorchScript decorators across the geometry module.

Politecnico di Milano & Bocconi AI and Neuroscience Association — Sep 2024 – Jun 2025
Computer Vision Lead, Intelligent Prosthetic Arm — Report

- Led 3D scene perception for an EEG-controlled prosthetic arm, combining object detection, monocular depth, segmentation, hand tracking, and Kalman-filtered state estimation.
- Implemented real-time grasp validation from object geometry and hand pose, with a GUI demonstrator and containerized deployment.

Accenture Jun 2025 – Sep 2025
Machine Learning Engineer

- Built a BERT/medical-LLM pipeline mapping diagnoses to a 20,000-class medical ontology with 94% accuracy; increased OCR/OpenCV document throughput 7.5×, from 2 to 15 documents/min.

TEACHING, LEADERSHIP & HONORS

Bocconi University Jan 2025 – Jul 2026

- Teaching Assistant, *Mathematical Analysis II*: selected by faculty to design and deliver three lectures on multivariable analysis and optimization.
- AI Tutor Developer, *Advanced Analysis and Optimization I*: developed the official course tutor for personalized guidance on ODE theory.
- Guest Speaker, *Bocconi Summer School*: delivered a computer-vision lecture and practical workshop.

Bocconi AI and Neuroscience Association Oct 2023 – Jul 2026

- Head of Research: coordinated and mentored 60 student researchers, defining ML projects with faculty and external research partners.

Bocconi Association for Mathematical Sciences Jul 2025 – Jul 2026

- Cofounder: established a faculty-mentored training program for the Imperial–Cambridge Mathematics Competition; placed 45th of 680 participants in round one and qualified for round two.

TECHNICAL SKILLS

Research: 3D/4D computer vision, multiple-view geometry, Gaussian Splatting, egocentric vision, uncertainty quantification.
Programming and ML: Python, PyTorch, NumPy, SciPy, OpenCV, Hugging Face.
Systems and tools: CUDA, COLMAP, Linux, Git, Docker, ROS2, Weights & Biases, GitHub Actions.

LANGUAGES

Italian native; **English** C2; **German** B2; **French** B2.